Thermodynamic description of the Al–Mg system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Al–Mg binary system as calculated with pycalphad from the SGTE dataset AlMg-12Tan.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Al) + β, lies at 723.9 K.

1. Introduction

The Al–Mg system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as intermetallic rich, peritectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of Y. Tang, Y. Du, L. Zhang, X. Yuan, G. Kaptay, Thermochim. Acta, 527, 131-42(2012). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

For Al, the liquidus meets the axis at 933.5 K. Pure Mg melts at 923.0 K.

2. Phase diagram and invariant reactions

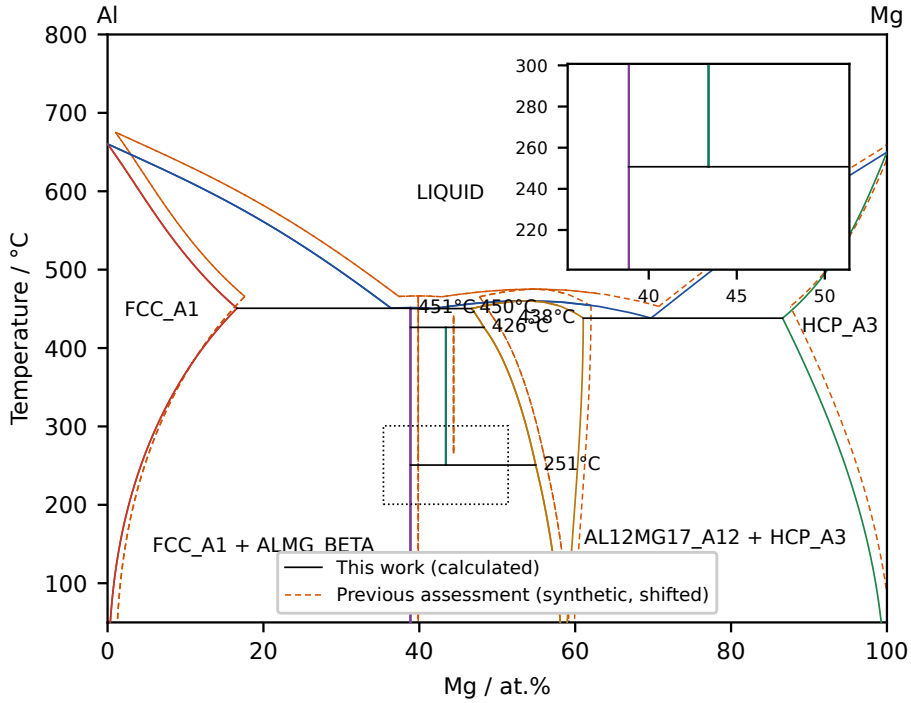


Fig. 1. Calculated Al–Mg phase diagram and a dashed, deliberately shifted second assessment (inset: enlarged view near the lowest invariant).

We obtain 723.9 K for the eutectic reaction (L → (Al) + β), where L has 36.4 at.% Mg. We obtain 723.5 K for the eutectic reaction (L → β + Al12Mg17), where L has 41.9 at.%. At 711.2 K the eutectic L → Al12Mg17 + (Mg) takes place; the L phase then contains 69.7 at.% Mg. The dataset places a peritectoid (β + Al12Mg17 → ε) at 699.5 K; the composition of ε is 43.4 at.% Mg. The eutectoid reaction ε → β + Al12Mg17 is calculated at 523.9 K, with ε at 43.4 at.%. Al12Mg17 melts congruently at 733.4 K and 53.5 at.%

Mg. β melts congruently at 724.7 K and 38.8 at.% Mg.

Table 1 compares the calculated temperatures with experimental values from the literature [3–5]; in this synthetic document those values and references are fictitious.

Table 1. Invariant reactions and congruent points, calculated and experimental (Al–Mg).

Reaction	Type	T (°C) this work	T exp.	Ref.	at.% Mg, 1	at.% Mg, 2	at.% Mg, 3
L → Al12Mg17	congruent melting	460.3	457.3	[5]	53.5	53.5	–
L → β	congruent melting	451.5	444.5	[3]	38.8	38.8	–
L → (Al) + β	eutectic	450.7	448.7	[5]	36.4	16.6	38.9
L → β + Al12Mg17	eutectic	450.4	456.4	[4]	41.9	38.9	46.7
L → Al12Mg17 + (Mg)	eutectic	438.1	–	–	69.7	61.0	86.6
β + Al12Mg17 → ε	peritectoid	426.4	–	–	38.9	48.3	43.4
ε → β + Al12Mg17	eutectoid	250.7	248.7	[5]	43.4	38.9	54.9

Experimental temperatures are fictitious values attributed to invented references [3–5] for testing evidence-class extraction; they are not data.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Al–Mg).

	Structure			Stability range / K		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	at.% Mg	at T / K
(Al)	cF4	A1	Cu	< 300	933.5	0.0–16.2	720
β	–	–	–	< 300	724.7	38.9	330
ε	–	–	–	523.9	699.5	43.4	680
Al12Mg17	cI58	A12	α-Mn	< 300	733.4	47.6–61.0	710
(Mg)	hP2	A3	Mg	< 300	923.0	86.7–100.0	710

The upper stability limit of ε is 699.5 K. The solubility of Al in (Mg) reaches 13.4 at.% at 711.2 K.

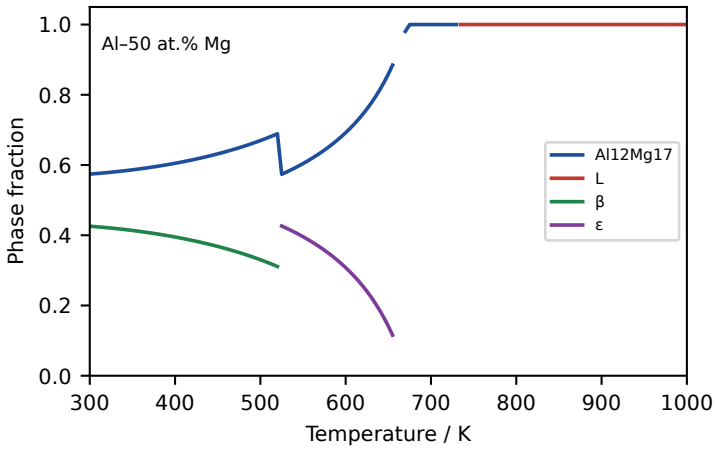


Fig. 2. Calculated equilibrium phase fractions versus temperature for Al–50 at.% Mg.

4. Thermodynamic properties

At 1000 K the integral mixing enthalpy of the liquid has its minimum, −2344 J/mol, at 52.0 at.% Mg. The calculated enthalpy of formation of ε at 298.15 K is −3.99 kJ/mol per mole of atoms. Its heat capacity at room temperature is 24.54 J/(mol K). For the equiatomic liquid at 1000 K, the Gibbs energy of mixing is −7.53 kJ/mol-atom.

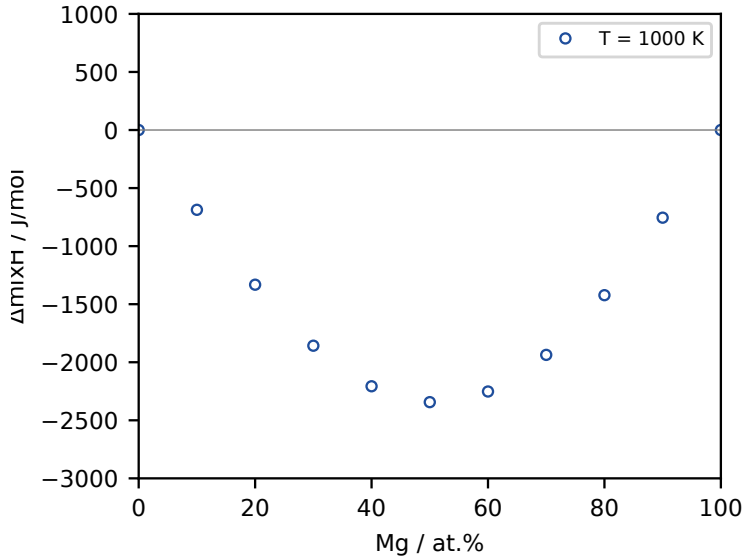


Fig. 3. Calculated mixing enthalpy of the liquid at 1000 K.

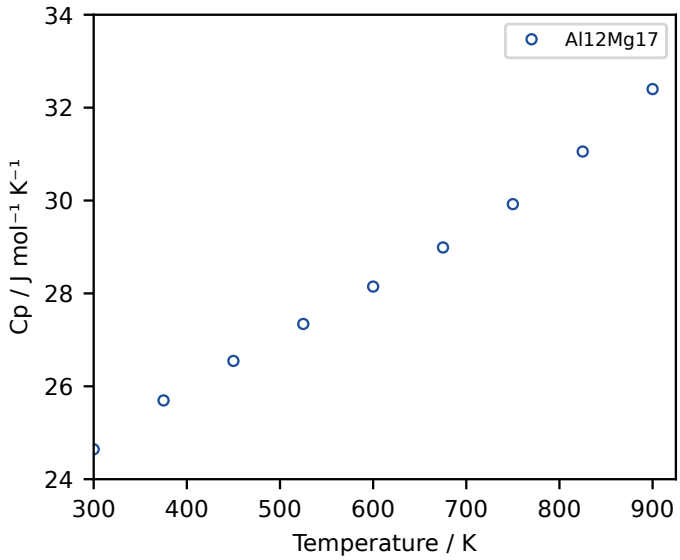


Fig. 4. Calculated heat capacity of Al12Mg17.

Table 3. Calculated formation properties of the intermediate phases at 298.15 K (Al–Mg).

Phase	at.% Mg	ΔfH(298.15 K) / kJ mol⁻¹	ΔfS / J mol⁻¹ K⁻¹	Cp(298.15 K) / J mol⁻¹ K⁻¹
Al12Mg17	56.3	−3.52	0.48	24.61
β	38.9	−4.53	−1.24	24.51
ε[a]	43.4	−3.99	−0.25	24.54

[a] Not stable at 298.15 K in this description; properties of the metastable phase.

Table 4. Integral and partial mixing properties of the liquid at 1000 K (Al–Mg).

at.% Mg	ΔmixH / kJ mol⁻¹	ΔmixG / kJ mol⁻¹	a(Al)	a(Mg)
10	−0.69	−3.19	0.9014	0.0548
20	−1.33	−5.14	0.7944	0.1144
30	−1.86	−6.46	0.6724	0.1893
40	−2.21	−7.25	0.5400	0.2845
50	−2.34	−7.53	0.4085	0.4001
60	−2.25	−7.29	0.2896	0.5301
70	−1.94	−6.52	0.1908	0.6634
80	−1.42	−5.20	0.1132	0.7887
90	−0.75	−3.24	0.0526	0.8995

Reference states: pure liquid Al and Mg at 1000 K; per mole of atoms.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] Y. Tang, Y. Du, L. Zhang, X. Yuan, G. Kaptay, Thermochim. Acta, 527, 131-42(2012).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.